A Machine Learning Based Heart Disease Prediction System Using ANN

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Abstract

With heart disease being among the top five causes of death across the globe, accurate and timely identification is one of the main healthcare challenges. The classification problem studied here is to predict the propensity of a patient to have heart disease based on their clinical and demographic features. If a physician develops early risk prediction, he can undertake preventative measures to improve patient care.

The primary goal of this research is to achieve the development of an intelligent prediction system that can classify heart disease cases with high accuracy using deep learning techniques.This study uses deep learning techniques. The goal of this study is to make diagnosis faster help doctors make decisions and find problems early on.

To do this the study uses a kind of computer program called an Artificial Neural Network model. This model is used to look at two possibilities and decide which one is correct. The model has layers that work together to understand the information and it uses something called ReLU and sigmoid to make decisions. The information was prepared in a way to help the model work better including handling missing information adjusting the importance of different factors and splitting the information into two groups, for testing and training.

The information used in this study is a set of numbers that are organized in a way.

Keywords:Heart Diseases;Artificial Neural Network;Deep Learning;Healthcare Prediction; Web Interface.

1 Introduction

1.1 Background of the Heart Disease Problem

Heart disease is a problem worldwide. It is one of the reasons people die. Heart disease affects how the heart and blood vessels work. Because peoples lifestyles are changing they are eating food getting stressed and not exercising enough. This means more and more people are getting heart disease. We need to find out who has heart disease early so we can help them. They can get better.

1.2 Factors that Increase the Risk of Heart Disease

There are things that can increase the risk of heart disease. These include how old you are, if you are a man or a woman if you have blood pressure, high cholesterol, diabetes if you are overweight if you smoke if you drink too much alcohol if someone in your family had heart disease if you are stressed and if you do not exercise enough. Often many of these things happen together. This makes it harder to figure out who will get heart disease. So we need to look at how all these things work to make an accurate diagnosis of heart disease.

1.3 How Big the Heart Disease Problem Is

The World Health Organization says that heart disease and other cardiovascular diseases cause about 17.9 million deaths every year. This is a lot of people. 32% Of all deaths worldwide. Many of these deaths happen because people do not get help enough or they do not take care of themselves. In some countries especially where people do not have a lot of money more and more people are getting heart disease because they do not have healthcare and they do not know how to stay healthy.

1.4 The Old Way of Diagnosing Heart Disease

doctors diagnose heart disease by looking at you doing blood tests looking at your ECG reports using imaging systems and using their experience. These methods are common. They can take a long time cost a lot of money and require a doctor who is an expert. Sometimes doctors can make mistakes. Take too long to decide what to do.

1.5 A New Way to Diagnose Heart Disease

This study suggests using a computer program called an Artificial Neural Network to predict if someone has heart disease. The Artificial Neural Network is like a computer brain that can learn from data. It uses information about the patient like how old they're

if they are a man or a woman their blood pressure, cholesterol and if they have chest pain to predict if they have heart disease.

1.6 What This Study Contributes

The thing this study does is make a smart and accurate heart disease prediction model using the Artificial Neural Network. This model can help doctors diagnose heart disease early and make decisions quickly. It can also be used to make a healthcare application for hospitals and clinics.

2 Background Study

2.1 Literature Review

Rahman and his team made a model that uses machine learning to predict heart disease. They used information from patients like their age, sex, cholesterol levels and blood pressure. First they fixed missing information. Made sure all the data was on the same scale. Then they tried methods like Logistic Regression, Decision Tree and Random Forest. The Random Forest method was the best getting it right 85% of the time.

Hossain and his team tried an approach. They picked the important information that affects heart disease and then used that to make predictions. They used methods like Support Vector Machine and K-Nearest Neighbors. Their study showed that picking the information makes the predictions more accurate getting it right about 88% of the time.

Islam and his team made a model that uses learning to predict heart disease. They used a lot of information from patients. Made sure it was clean and ready to use. Their model had layers and used a special function to help it learn. The model was very good getting it right 90% of the time. This shows that deep learning is really good at finding patterns in medical information.

Khan and his team made their model better. They used techniques to make sure their model was working as well as it could. They tried different options to find the best one. Their model used information from patients. Made sure it was all on the same scale. The new model was the best getting it right 92%time. It was even better than the models that use machine learning. Heart disease prediction using these models is very good. The models like Random Forest and deep learning models are very helpful, for heart disease prediction.

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Title	Number of Data	Parameters Used	Methodology	Accuracy
ML-Based Heart Disease Prediction	300	13 Features	Logistic Regression, RF	85%
Hybrid Feature Selection Model	500	Selected Features	SVM, KNN	88%
ANN-Based Heart Prediction	700	Clinical Data	ANN Model	90%
Optimized ANN Model	1000	Medical Features	ANN + Hyperparameter Tuning	92%

Table 1: Summary of Literature Review

3 Materials & Method

3.1 Proposed Methodology

In this study, a dataset is used to train a neural network model. The following steps are followed:

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1. Data Collection
2. Data Preprocessing (Normalization)
3. Model Training
4. Testing and Evaluation

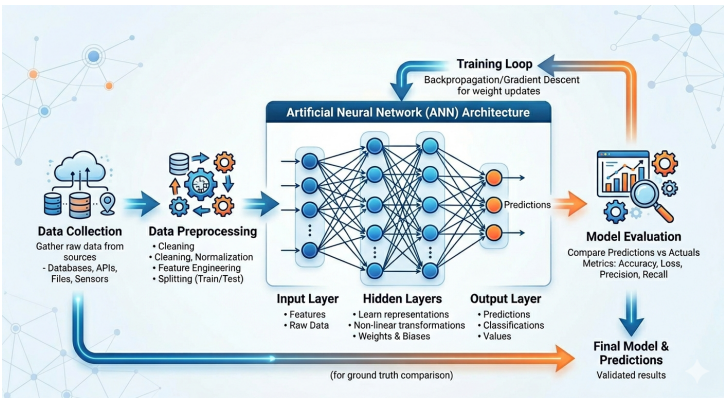

Figure 1: ANN Methodology Diagram

3.2 Dataset

We used a dataset with medical information related to heart disease. This dataset has 541 entries. Includes details, like age, sex, blood pressure, cholesterol level and chest pain type. The dataset shows if a patient has heart disease or not. It has clinical features. The entries are marked as having heart disease (1) or not (0)

Age	Sex	BP	Cholesterol	Target
52	1	125	212	1
45	0	130	180	0
60	1	140	240	1
38	0	120	190	0

Table 2: Sample Data Representation

3.3 Data Processing

Data preprocessing is an important step. It helps make the model better. We did things to preprocess the data:

* Handling Missing Values: We replaced missing values with the average value. This way we had data.

* Feature Scaling: We used a tool called StandardScaler to make all features equal. This helps the model learn better.

* Train-Test Split: We split the data into two parts. 80% Was for training. 20% Was for testing. This helps us see how well the model works.

* Data Cleaning: We removed data that was not useful or had errors. This makes the model more accurate.

These steps helped us get a dataset. It was clean and ready, for our Artificial Neural Network model. The data was also normalized. This helps the model learn from the data. We did all this to make sure our Artificial Neural Network model works.

3.4 Machine Learning/Deep Learning Models

3.4.1 Proposed Artificial Neural Network

The main part of the system we are talking about is a kind of Artificial Neural Network. This Artificial Neural Network is designed to look at information that is arranged in tables. It is really good at figuring out what things mean in this kind of data. Some other kinds of deep learning models like the ones that look at pictures are great at finding patterns in images.. When we are looking at medical information like blood pressure or cholesterol levels or heart rate the Artificial Neural Network is really good at finding connections, between these things that are not simple or straightforward.

3.4.2 Fine Tuning Process Models

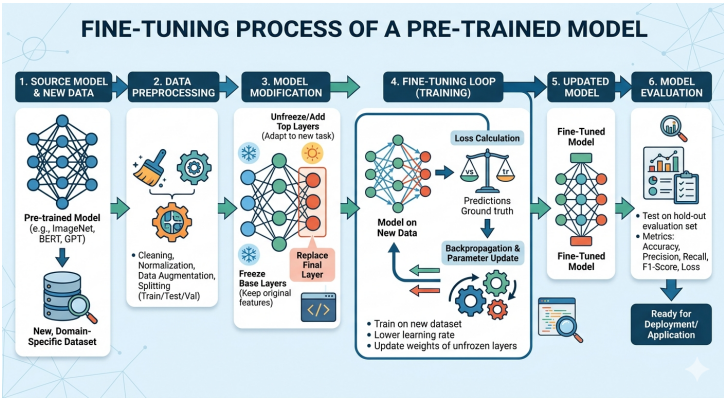

Figure 2: ANN Methodology Diagram

4 Results and Discussion

4.1 Experimental Setup

In this experiment, an investigation was carried out based on a heart disease data set that had several clinical attributes such as age, gender, blood pressure, cholesterol level, and maximum heart rate. Prior to training, preprocessing of the data set was done to address the issue of missing data and normalization of the attribute values in order to enhance the efficiency of the model.

For the purposes of evaluating the model's effectiveness, the data set was partitioned into training and testing datasets. The 5-fold cross-validation method was employed to make sure that the model is robust and generalizable. This helps to minimize any biases and give a more realistic estimate of the performance of the model.

Artificial Neural Network (ANN) was used in this experiment for classification. The model comprises an input layer, one or more hidden layers, and an output layer. In the course of training, model parameters were tuned through backpropagation and optimizer.

Model evaluation was performed by looking at accuracy, loss, confusion matrix, and ROC curve.

4.2 Confusion Metrix

To measure the effectiveness of the classifier, a confusion matrix was utilized to compare the actual and predicted values. The confusion matrix shows details about the correct predictions for positive and negative classes, which are heart disease cases and non-heart disease cases, respectively, as well as the incorrect predictions made by the classifier.

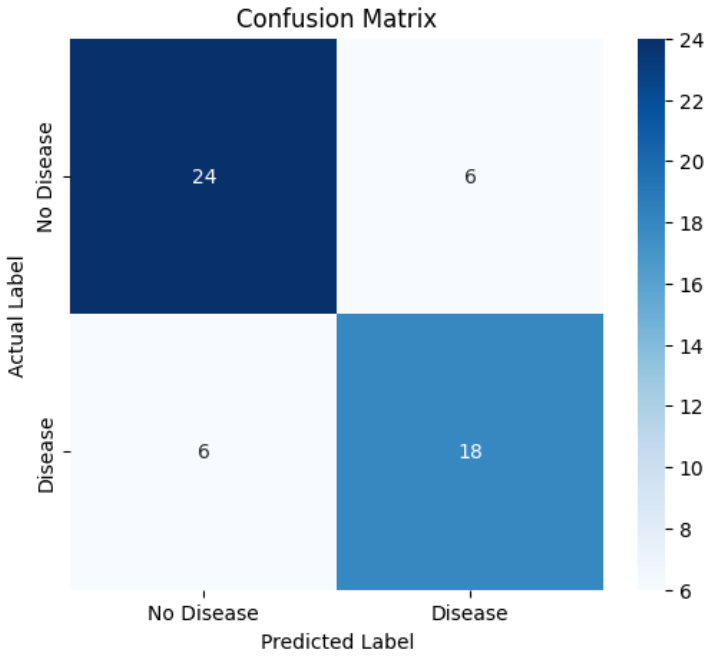

Figure 3: Confusion Metrix

4.3 Result Analysis

The model performance is evaluated using accuracy. Higher accuracy indicates better performance.

- Training Accuracy: 100%
- Validation Accuracy: 73% - 75%

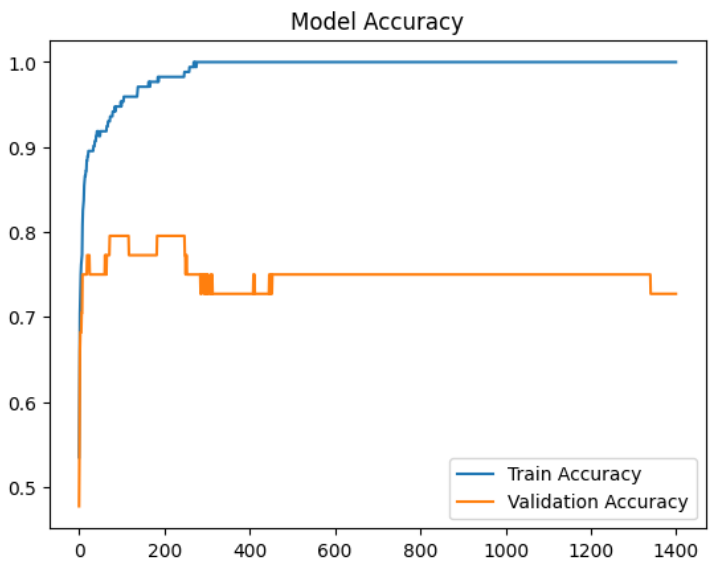

Figure 4: Model Accuracy

5 Conclusion

This study is about using Artificial Neural Network to predict heart disease. The Artificial Neural Network is based on clinical data. We used information like age, sex, blood pressure and cholesterol level to develop the model. We made sure to clean up the data by handling missing values and scaling the features. We also split the data into sets to make the model work better. The Artificial Neural Network has layers that help it learn from the data. It can find relationships between the different features. Our results show that the model is good at predicting heart disease.

There are some problems with this study. The data we used is not very big. May not have all kinds of patients. This can affect how well the model works. Also the model might not work well if some important medical features are missing. Another issue is that Artificial Neural Network models can be hard to understand. This is a problem when it comes to healthcare because we need to know how the model is making decisions.

For studies we can make some improvements. We can test the model on more diverse data to make it work better for all patients. We can also try using types of models or combining Artificial Neural Network with other techniques. We can pick the important features and adjust the model to make it work better. It would be great to make the model part of a real-time system that healthcare professionals and patients can use. This could be a website or a mobile app. The Artificial Neural Network can be very helpful, in predicting heart disease if we make these improvements..

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